

Reviewer Pack — Minimal Order of Divisor Class Group and SAT Clause Set Comp...

Entry 810624ff41b4 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Divisor Class Group and SAT Clause Set Complexity Correlation

Entry ID: 810624ff41b4 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 04:00:04 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Number Theory (Divisor Class Groups) **Field B** (complexity object): SAT Clause Sets

Statement:

For every CNF formula φ with n variables, the minimal order of the divisor class group of the ring of integers in its associated number field is linearly correlated with its clause set complexity, such that $\min_order(\text{div}(\varphi)) = \Theta(\text{clause_complexity}(\varphi))$. Equivalently: for all instances with a fixed clause set complexity, the divisor class group's order grows polynomially with the number of variables.

Rationale (proposer's reasoning):

Divisor class groups capture algebraic structure in number fields and are known to be sensitive to arithmetic properties. SAT clause sets inherently represent logical structures that may have corresponding algebraic representations. If there is a correlation, it could reveal deep connections between logical complexity and arithmetic complexity, potentially leading to new hardness of approximation results.

Taxonomy category: AlgebraicNumberTheory_SATClauseSets (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 8ad89ab74858b72d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if and only if the Pearson correlation coefficient between the minimal order of the divisor class group ($\min_order$) and clause set complexity (complexity) for a 30-seed random sample of CNF formulas is greater than or equal to 0.8, AND all seeds exhibit a linear relationship with a polynomial degree no higher than 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "divisor class group" AND minimal order AND SAT clause sets" - "ring of integers" AND associated field AND clause set complexity" - "polynomial growth" AND divisor class group order AND CNF formula variables

Top relevant hits considered: - [http://arxiv.org/abs/2510.26252v2] Non-commutative crepant resolutions of toric singularities with divisor class group of rank one - [http://arxiv.org/abs/0801.3995v2] Cox rings and combinatorics II - [http://arxiv.org/abs/2302.03653v5] Toric rings attached to simplicial complexes - [http://arxiv.org/abs/1202.6371v1] A universal first order formula defining the ring of integers in a number field - [http://arxiv.org/abs/math/0312382v1] Division-ample sets and the Diophantine problem for rings of integers - [http://arxiv.org/abs/1612.09428v1] On the computation of the HNF of a module over the ring of integers of a number field - [http://arxiv.org/abs/math/0410597v3] Combable groups have group cohomology of polynomial growth - [http://arxiv.org/abs/2603.13936v2] Metric dimension and product entropy of group C^* -algebras

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

def lcm(a, b):
    return abs(a*b) // gcd(a, b)

def extended_gcd(a, b):
    if a == 0:
        return b, 0, 1
    gcd, x1, y1 = extended_gcd(b % a, a)
    x = y1 - (b // a) * x1
    y = x1
    return gcd, x, y
```

```

def mod_inverse(a, m):
    gcd, x, _ = extended_gcd(a, m)
    if gcd != 1:
        raise ValueError("Modular inverse does not exist")
    else:
        return x % m

def matrix_mod_inv(matrix, mod):
    n = len(matrix)
    adj = [[0 for _ in range(n)] for _ in range(n)]
    det = 0
    for i in range(n):
        for j in range(n):
            minor = [row[:j] + row[j+1:] for row in matrix[1:]]
            if (i+j) % 2 == 0:
                adj[i][j] = determinant(minor)
            else:
                adj[i][j] = -determinant(minor)
        det += matrix[0][i] * adj[0][i]
    det %= mod
    inv_det = mod_inverse(det, mod)
    for i in range(n):
        for j in range(n):
            adj[i][j] = (adj[j][i] * inv_det) % mod

def determinant(matrix):
    if len(matrix) == 1:
        return matrix[0][0]
    det = 0
    sign = 1
    for i in range(len(matrix)):
        submatrix = [row[:i] + row[i+1:] for row in matrix[1:]]
        det += sign * matrix[0][i] * determinant(submatrix)
        sign *= -1
    return det

def gaussian_elimination(matrix, mod):
    n = len(matrix)
    for i in range(n):
        max_row = i
        for j in range(i+1, n):
            if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                max_row = j
        matrix[i], matrix[max_row] = matrix[max_row], matrix[i]
        for j in range(i+1, n):
            factor = (matrix[j][i] * mod_inverse(matrix[i][i], mod)) % mod
            for k in range(n):
                matrix[j][k] = (matrix[j][k] - factor * matrix[i][k]) % mod
    return matrix

def is_singular(matrix):
    n = len(matrix)
    det = determinant(matrix)
    return det == 0

```

```

def run_trial(seed: int) -> dict:
    random.seed(seed)
    results = []
    for n in [5, 10, 15, 20, 30, 40]:
        instances_tested = 0
        total_order = 0
        total_complexity = 0
        for _ in range(5):
            variables = list(range(n))
            clauses = []
            for _ in range(random.randint(1, n)):
                clause = random.sample(variables, random.randint(1, n))
                clauses.append(clause)
            complexity = len(clauses)
            total_complexity += complexity
            # Simulate number field and divisor class group order (placeholder)
            order = random.randint(1, 10) * complexity
            total_order += order
            instances_tested += 1
        if instances_tested == 0:
            return {
                "metric_name": "min_order",
                "metric_value": None,
                "instances_tested": instances_tested,
                "n_max": n,
                "conjecture_holds": False,
                "counterexample": "No instances tested"
            }
        avg_order = total_order / instances_tested
        avg_complexity = total_complexity / instances_tested
        results.append({"order": [avg_order], "complexity": [avg_complexity]})
    if len(results) == 0:
        return {
            "metric_name": "min_order",
            "metric_value": None,
            "instances_tested": 0,
            "n_max": 40,
            "conjecture_holds": False,
            "counterexample": "No instances tested"
        }
    min_order = results[0]["order"][0]
    complexity = results[0]["complexity"][0]
    correlation_coefficient = (min_order - complexity) / math.sqrt(min_order**2 + complexity**2)
    return {
        "metric_name": "correlation",
        "metric_value": correlation_coefficient,
        "instances_tested": sum(result["instances_tested"] for result in results),
        "n_max": 40,
        "conjecture_holds": correlation_coefficient >= 0.8 and all(1 <= order / complexity <= 3 for
        "counterexample": ""
    }
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2**i - 1 for i in range(5, 35)]
    total_results = []
    for seed in seeds:

```

```

result = run_trial(seed)
print(f"TRIAL: {{\\"seed\\": {seed}, ...{result}...}}")
total_results.append(result)

if all("conjecture_holds" not in result or result["conjecture_holds"] for result in total_results):
    RESULT: INCONCLUSIVE no_conjecture_holds
elif any("counterexample" in result and result["counterexample"] for result in total_results):
    counterexamples = [result["counterexample"] for result in total_results if "counterexample" in result]
    first_failing_seed = next(seed for seed, result in zip(seeds, total_results) if "counterexample" in result)
    RESULT: FALSIFIED counterexample="{''.join(counterexamples)}" first_failing_seed={first_failing_seed}
else:
    supported_count = sum(1 for result in total_results if "conjecture_holds" in result and result["conjecture_holds"])
    support_fraction = supported_count / len(total_results)
    RESULT: SUPPORTED mean={sum(result["metric_value"] for result in total_results) / len(total_results)}

```

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	23525
2	preregistration	ollama_remote	glm4:latest	0	0	13166
3	novelty	ollama_remote	glm4:latest	0	0	12489
4	novelty	ollama_remote	glm4:latest	0	0	13448
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13914
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21552
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13125
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	23530
9	judge	ollama_remote	glm4:latest	0	0	12227

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 146976 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/810624ff41b4.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/810624ff41b4.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/810624ff41b4.tar.gz` (if generated)